

EXPERIMENT 2 – Illustrate UNIX Commands and Shell Programming

Aim: To study and execute UNIX commands and write shell programs.

2.1 UNIX Commands

Key commands covered: `date`, `cal`, `echo`, `who`, `ls`, `mkdir`, `cd`, `rmdir`, `cat`, `cp`, `mv`, `rm`, `wc`, `head`, `tail`, `grep`, `sort`, `pipe ()`, `tr`

2.2 Shell Programming

Program 1 – Greatest Among Three Numbers

Aim: To find the greatest among three given numbers.

```
#!/bin/bash

echo "ENTER THREE NUMBERS"
read a b c

if [ $a -gt $b ] && [ $a -gt $c ]
then
    echo "$a is greater"
elif [ $b -gt $c ]
then
    echo "$b is greater"
else
    echo "$c is greater"
fi
```

Output (Input: 10 25 7):

```
ENTER THREE NUMBERS
25 is greater
```

Program 2 – Factorial of a Given Number

Aim: To find the factorial of a given number.

```
#!/bin/bash

echo "ENTER THE NUMBER:"
read n

fact=1

while [ $n -gt 1 ]
do
    fact=$((fact * n))
    n=$((n - 1))
done

echo "FACTORIAL OF THE GIVEN NUMBER IS $fact"
```

Output (Input: 5):

```
ENTER THE NUMBER:
FACTORIAL OF THE GIVEN NUMBER IS 120
```

Program 3 – Sum of Odd Numbers up to N

Aim: To find the sum of odd numbers up to N.

```
#!/bin/bash

echo "ENTER THE RANGE:"
read n

x=1
sum=0

while [ $x -le $n ]
do
    sum=$((sum + x))
    x=$((x + 2))
done
```

```
echo "SUM = $sum"
```

Output (Input: 10):

```
ENTER THE RANGE:  
SUM = 25
```

Program 4 – Generation of Fibonacci Numbers

Aim: To generate Fibonacci numbers up to a given limit.

```
#!/bin/bash  
  
echo "ENTER THE LIMIT:"  
read n  
  
p=-1  
q=1  
i=1  
  
while [ $i -le $n ]  
do  
    r=$((p + q))  
    p=$q  
    q=$r  
    echo "$r"  
    i=$((i + 1))  
done
```

Output (Input: 8):

```
ENTER THE LIMIT:  
0  
1  
1  
2  
3  
5  
8  
13
```

Program 5 – Arithmetic Calculator

Aim: To implement an arithmetic calculator using the case statement.

```
#!/bin/bash  
  
echo "ENTER THE VALUE OF A:"  
read a  
echo "ENTER THE VALUE OF B:"  
read b  
echo "ENTER THE OPTION TO PERFORM"  
echo "1. ADDITION"  
echo "2. SUBTRACTION"  
echo "3. MULTIPLICATION"  
echo "4. DIVISION"  
read op  
  
case "$op" in  
    1) echo "Result = $((a + b))" ;;  
    2) echo "Result = $((a - b))" ;;  
    3) echo "Result = $((a * b))" ;;  
    4) echo "Result = $((a / b))" ;;  
    *) echo "Invalid Option" ;;  
esac
```

Output (Input: A=12, B=4, Option=1 Addition):

```
ENTER THE VALUE OF A:  
ENTER THE VALUE OF B:  
ENTER THE OPTION TO PERFORM  
1. ADDITION  
2. SUBTRACTION  
3. MULTIPLICATION  
4. DIVISION  
Result = 16
```

Program 6 – Largest Digit of a Number

Aim: To find the largest digit in a given number.

```
#!/bin/bash
```

```

echo "ENTER THE NUMBER"
read a

max=0

while [ $a -gt 0 ]
do
    r=$((a % 10))
    if [ $r -gt $max ]
    then
        max=$r
    fi
    a=$((a / 10))
done

echo "THE LARGEST DIGIT OF THE NUMBER: $max"

```

Output (Input: 9237):
ENTER THE NUMBER
THE LARGEST DIGIT OF THE NUMBER: 9

Program 7 – Palindrome String Check

Aim: To check whether a given string is a palindrome.

```

#!/bin/bash

echo "ENTER THE STRING TO CHECK PALINDROME"
read str

len=$(echo -n "$str" | wc -c)
i=1
j=$((len / 2))

while [ $i -le $j ]
do
    k=$(echo "$str" | cut -c $i)
    l=$(echo "$str" | cut -c $len)
    if [ "$k" != "$l" ]
    then
        echo "$str is not a palindrome"
        exit
    fi
    i=$((i + 1))
    len=$((len - 1))
done

echo "$str is a palindrome"

```

Output (Input: "madam"):
ENTER THE STRING TO CHECK PALINDROME
madam is a palindrome

Output (Input: "hello"):
ENTER THE STRING TO CHECK PALINDROME
hello is not a palindrome

Program 8 – Reverse of a Given Number

Aim: To find the reverse of a given number.

```

#!/bin/bash

echo "ENTER THE NUMBER"
read n

rnum=0

while [ $n -ne 0 ]
do
    remainder=$((n % 10))
    rnum=$((rnum * 10 + remainder))
    n=$((n / 10))
done

echo "REVERSE OF THE NUMBER IS $rnum"

```

Output (Input: 12345):
ENTER THE NUMBER
REVERSE OF THE NUMBER IS 54321

Result: The shell programs were executed successfully and outputs verified.
